

René MOURAMBOU DIYEMA

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#) | [Medium](#) | [LeetCode](#) | [HackerRank](#) | Paris, France

I'm a **Quantitative Analyst** with an **MSc in Probability, Statistics and Finance** from Gustave Eiffel University. I **develop, calibrate and validate stochastic models for equity and fixed-income derivative pricing and risk**, turning market data into prices, hedges and risk measures through stochastic calculus, Monte Carlo methods and statistical inference. I am seeking a **Quantitative Finance internship starting in April 2027**.

EDUCATION

MSc Probability, Statistics and Finance — Gustave Eiffel University, France Courses: High-dimensional statistics, simulations and copulas, empirical estimation and extreme values, deterministic and stochastic approximation, stochastic calculus, fractional models in finance, statistical learning, derivatives, high-frequency data and order books, interest rates and credit	Sept 2026 – Sept 2027
MSc Artificial Intelligence — Paris-Saclay University, France Courses: Probabilistic generative models, signal processing, frugal AI, theory and advanced machine learning, advanced optimization	Sept 2025 – Sept 2026
MSc Mathematics, Vision, Learning — ENS Paris-Saclay, France Courses: Stochastic calculus, statistical learning with extreme values, time series analysis, computational statistics, convex optimization	Sept 2024 – Sept 2025
MSc Applied Mathematics — ESIEE Institute of Technology, France (gap year 2023–2024) Courses: Probability for finance, discrete Markov chains, stochastic calculus, financial time series, jump models, risk measures, financial markets, Fixed Income, Equities, Portfolio Management, Interest Rates	Sept 2022 – Sept 2024
BSc Applied Mathematics — Sup'Management Engineering School, Gabon Courses: Linear algebra, calculus, probability and statistics, Markov processes, optimization	Sept 2017 – Sept 2019

PROFESSIONAL EXPERIENCE

Machine Learning Researcher Full-Time Internship Nokia, France • Structured large-scale time-series signals for statistical estimation , controlling data quality and consistency across sources. • Characterised the series as stochastic processes , estimating temporal dependence , stationarity and distributional properties . • Applied statistical signal processing (filtering, spectral analysis) to isolate the components used for model estimation. • Specified and estimated temporal models (CNN, ResNet, Transformer) under supervised, semi-supervised and self-supervised regimes. • Benchmarked out-of-sample performance through hypothesis testing against reliability and generalization criteria. • Ran large-scale numerical experiments on GPU , comparing specifications and tracking estimation results across trials. • Documented model assumptions and estimation procedures , and presented results to business and engineering teams to guide decisions. Technologies: Python, NumPy, SciPy, PyTorch, Time Series, Stochastic Processes, Signal Processing, Statistical Inference, Hypothesis Testing, CUDA	Apr 2026 – Sep 2026
Machine Learning Researcher Full-Time Internship SEGULA Technologies, France • Prepared high-dimensional datasets and constructed informative features to support forecasting . • Developed reduced-order models for multi-physics systems governed by nonlinear PDEs . • Implemented Galerkin projection with error control and numerical convergence for finite- and infinite-order dynamics. • Modelled system dynamics with Autoencoders, LSTM and GNNs to produce multi-step forecasts. • Quantified forecast uncertainty and stability through residual analysis , refining specifications to reduce prediction error. • Tracked forecast performance over time to detect drift and recalibrate model parameters. Technologies: Python, NumPy, SciPy, PyTorch, Nonlinear PDEs, Galerkin, Reduced-Order Modeling, Forecasting, LSTM, GNNs, CUDA	Apr 2025 – Sep 2025
Machine Learning Researcher Full-Time Internship Evolukid, France • Structured and annotated multimodal datasets to support probabilistic evaluation . • Estimated probabilistic models for classification and detection , combining text and image signals for decision-making. • Evaluated decision rules via controlled A/B experiments in a real-time decision loop , measuring latency and decision quality . Technologies: Python, PyTorch, Pandas, Probabilistic Models, A/B Testing, Statistical Evaluation, Decision Models	Jan 2024 – Jun 2024

PROJECTS

Exotic Derivatives: Autocallable Pricing, Hedging and Model Risk • Built an arbitrage-free implied volatility surface from S&P 500 options and derived Dupire local volatility . • Built Heston and local-stochastic volatility models with a leverage function on the same surface. • Valued a 3-year autocallable and a barrier reverse convertible by Monte Carlo under each model for model risk . • Computed Greeks by bumping, simulated weekly delta hedging and decomposed profit and loss by Greek. Technologies: Python, QuantLib, Monte Carlo, Implied and Local Volatility, Heston, Local-Stochastic Volatility, Autocallables, Delta Hedging	Sept 2026 – Oct 2026
Rough Volatility: Calibration and Monte Carlo Pricing under Rough Bergomi • Estimated the Hurst exponent from realized volatility series to characterise rough volatility dynamics. • Simulated fractional Brownian motion with the hybrid scheme to implement the rough Bergomi model. • Priced European options by Monte Carlo with control variates and antithetic variance reduction . • Calibrated on a real implied volatility surface and compared delta-hedging P&L to Black-Scholes and Heston . Technologies: Python, C++, SciPy, QuantLib, Stochastic Calculus, Monte Carlo, Fractional Brownian Motion, Calibration, Implied Volatility, Greeks	Apr 2026 – May 2026
Portfolio Risk Engine: Extreme Value Theory, Copulas and Expected Shortfall • Modelled marginal tail risk with GARCH and Generalized Pareto distributions (Peaks-Over-Threshold). • Modelled dependence with Student-t, Clayton and Gumbel copulas , and estimated tail dependence . • Computed VaR and Expected Shortfall by historical, parametric and copula Monte Carlo , with Ledoit-Wolf covariance. • Backtested models with Kupiec, Christoffersen and Acerbi-Székely tests to quantify model risk . Technologies: Python, NumPy, SciPy, Statsmodels, arch, Scikit-learn, EVT, Copulas, GARCH, VaR, Expected Shortfall, Backtesting	Dec 2025 – Jan 2026

SKILLS

- **Languages and tools:** Python, C++, SQL, Linux, Git, Docker, Kubernetes
- **Libraries:** QuantLib, Statsmodels, CVXPY, NumPy, SciPy, Pandas, Matplotlib, Seaborn, PyTorch, TensorFlow, Keras, Scikit-learn
- **Quantitative Finance:** Stochastic Calculus, Monte Carlo, Derivative Pricing, Greeks, VaR, Expected Shortfall, Calibration, Implied Volatility
- **Probability & Statistics:** Stochastic Processes, Statistical Inference, Hypothesis Testing, Time Series, Regression, PCA, Covariance Estimation